THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** asexual reproduction	clones
2	**(a)** tubers (eyes)	underground stem
3	**(b)** spores in the air settle on moist bread	warm and moist
4	**(d)** 23	46 chromosomes in total
5	**(c)** anther	male gametes
6	**(c)** cross-pollination	genetic variation
7	**(b)** water	aquatic plant
8	**(a)** egg (ovum)	produced in the ovary
9	**(d)** the egg is not fertilised	lining sheds
10	**(d)** trimesters	each about three months

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The process by which living beings produce new individuals of their own kind (1); it continues the species (1).	1+1
12	Asexual: one parent, identical offspring; sexual: two parents, offspring with variations (1+1).	1+1
13	New plants arise from vegetative parts (1); potato tuber and Bryophyllum leaf (1).	1+1
14	A small outgrowth on the parent that detaches as a new individual (1); yeast or hydra (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A small bud forms on the parent cell (1); the nucleus divides and enters the bud (1); the bud detaches and grows into a new yeast cell (1).	1+1+1
16	By spores (1); keep moist bread in a warm, humid, dark place; spores from the air germinate in 3 days (2).	1+2
17	Cutting, grafting, layering, tissue culture (any three) (2); quick production of identical plants with desired traits (1).	2+1
	OR: Plants produced are identical to the parent (1.5); they flower and fruit earlier and keep desirable characters (1.5).	1½+1½
18	Advantages: quick, single parent, rapid population growth (1.5); disadvantage: no variation, so poor adaptation to change (1.5).	1½+1½
	OR: Fusion of gametes from two parents (1); random combinations of chromosomes (1); variation helps survival and evolution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Producing new individuals of one's own kind (1). (ii) Number of parents, variation, speed, examples (3). (iii) Mitosis copies chromosomes exactly (1).	1+3+1
	OR: (i) Hydra/yeast; Rhizopus; potato/Bryophyllum (3). (ii) Single parent and identical offspring (2).	3+2
20	(i) Bread mould (Rhizopus) (1). (ii) Spores in the air (1). (iii) By spore formation (1). (iv) Penicillin antibiotic; decomposing wastes (2).	1+1+1+2
	OR: (i) A piece of stem with nodes is planted in moist soil (2). (ii) Joining a shoot to a rooted stock (1). (iii) Faster growth, identical plants, seedless varieties (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 12: **Reproduction: How Life Continues** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** identical to the parent	clones
2	**(c)** plantlets on leaf margins	vegetative propagation
3	**(d)** Low temperature slows down microbial growth	fungi need warmth
4	**(b)** 23	haploid
5	**(a)** pistil	stigma, style, ovary
6	**(c)** insects	attract pollinators
7	**(d)** fish and frogs	in water
8	**(a)** zygote	fertilisation
9	**(a)** 28 days	21–35 days
10	**(d)** condom	stops sperm reaching the egg

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Light, single-celled reproductive units (1); to increase the chance of reaching suitable places (1).	1+1
12	Showed that life comes only from pre-existing life, not from non-living matter (2).	2
13	An offspring genetically identical to its parent (2).	2
14	Thread-like structures in the nucleus carrying genetic information (1); 46 (23 pairs) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stamen: filament and anther (1.5); pistil: stigma, style, ovary (1.5).	1½+1½
16	Wrap flower buds in muslin, with different treatments (1); remove stamens in one, hand-pollinate others (1); fruit forms except where the stamens were removed (1).	1+1+1
17	Self: pollen from the same flower or plant; cross: from a flower of a different plant (1.5); cross-pollination gives more variation (1.5).	1½+1½
	OR: Insects: bright, scented, nectar, sticky pollen; wind: small, dull, light dry pollen in large amounts (1.5+1.5).	1½+1½
18	Pollen tube reaches the ovule and the male gamete fuses with the egg (1); zygote becomes the embryo; ovule becomes the seed (1); ovary becomes the fruit (1).	1+1+1
	OR: Wind (dandelion), water (coconut), animals (burs) (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Diagram (2). (ii) Sepals protect, petals attract, stamens make pollen, pistil contains ovules (2). (iii) Unisexual: only stamens or pistils; bisexual: both (1).	2+2+1
	OR: (i) Cross-pollination (1). (ii) Wind (1). (iii) Gives variation and hybrid vigour (1). (iv) Anther and stigma (2).	1+1+1+2
20	(i) Self and cross (1). (ii) Wind, water, insects, birds with flower features (2). (iii) Pollen tube grows, fertilisation, seed and fruit form (2).	1+2+2
	OR: (i) Most pollen is lost (1). (ii) Sticky or spiny, large (1). (iii) To attract pollinators (1). (iv) Coral tree or hibiscus (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.